

PROFESSIONAL SUMMARY

Computational biologist and PhD candidate in Biomedical Engineering at NYU, developing multi-omic frameworks (RNA-seq, proteomics, metabolomics, methylomics, WGS/WES) to identify translational biomarkers of cardiovascular disease. Applies machine learning to large clinical cohorts, including the ISCHEMIA trials biorepository, to characterize platelet transcriptomics, clonal hematopoiesis, and autoimmune-cardiovascular intersections. Author of 5+ first-author and 20+ contributing peer-reviewed publications, with two pending U.S. provisional patents.

EDUCATION

- **PhD, Biomedical Engineering**, NYU Tandon School of Engineering 2023–2026, New York City, NY
- **Master of Science, Biomedical Informatics**, NYU Grossman School of Medicine 2022–2023, New York City, NY
- **Bachelor of Arts, Chemistry**, Reed College 2014–2018, Portland, OR

PROFESSIONAL EXPERIENCE

- **Graduate Research Assistant**, New York University 2023–2026, New York, NY
 - Built and validated a whole-blood multiomic risk model for chronic coronary disease, integrating transcriptomic (n=646) and methylomic (n=732) profiling validated across two independent external cohorts (*Nature Communications*, 2026).
 - Characterized clonal hematopoiesis (CHIP) prevalence (23%) and its relationship to cardiovascular risk in 895 ISCHEMIA trial participants via whole-exome sequencing (*Circulation: Genomic and Precision Medicine*, 2025).
 - Developed and validated a platelet gene expression score (SLAP-GES, $r=0.76$, $P<0.001$) linking systemic lupus disease activity to vascular dysfunction (*JACC: Basic to Translational Science*, 2025).
 - Co-developed the Platelet Reactivity Expression Score (PRESS), an RNA-sequencing tool with an AUC of 0.81 for identifying hyperreactive platelet phenotypes; high-scoring patients had a 90% higher risk of cardiovascular events (*Nature Communications*, 2024).
- **Research Associate**, Oregon Health & Science University 2021–2022, Portland, OR
 - Led pre-clinical studies on catheter-based ultrasound as therapy for peripheral artery disease and pulmonary embolism; contributed to 8+ peer-reviewed publications.
 - First-authored a book chapter on Ultrasound Imaging in Inflammation Research.
- **Research Assistant 2**, Oregon Health & Science University 2018–2021, Portland, OR
 - Characterized vascular and hemodynamic effects of endovascular ultrasound using contrast echocardiography, histology, and molecular imaging.
 - Investigated molecular mechanisms of aortic stenosis as potential therapeutic targets.
- **Volunteer Researcher, Addiction Medicine Echo**, Oregon Health & Science University 2019–2020, Portland, OR
 - Performed qualitative thematic analysis on Echo program participation; wrote clinical notes summarizing case presentations.
- **Project Coordinator, Clinical Research Investigative Studies Program**, Oregon Health & Science University 2016–2018, Portland, OR
 - Managed patient enrollment across emergency department studies; coordinated Quickbrain MRI Study as Project Lead; trained new volunteers.
- **Advisory Board Member**, International Partners 2018–2022, Silver Spring, MD
 - Facilitated strategic planning and wrote grant proposals to fund community-driven engagement programs for youth with disabilities.

SKILLS AND TECHNIQUES

- **Bioinformatics**: RNA-seq, scRNA-seq, WGS/WES, proteomics, methylomics, metabolomics, GWAS; DESeq2, Scanpy, GATK.
- **Data Science**: Machine learning (Scikit-learn, TensorFlow, PyTorch), Docker, Git.
- **Cloud Computing**: AWS, GCP, HPC/Slurm.
- **Languages**: Python, R, Bash, L^AT_EX, SQL, Matlab.

AWARDS

- **2022:** ASE Foundation Early Career Investigator. *ASE Scientific Sessions 2022, Seattle, WA*
- **2022:** Alan S. Pearlman Research Award. *ASE Scientific Sessions 2022, Seattle, WA*

PATENTS

- **2025:** A Circulating Protein Signature of Platelet Reactivity Predicts Cardiovascular Risk. *U.S. Provisional Patent Application No. 63/912,729*, filed November 6, 2025. With J.S. Berger, T.J. Barrett. Assignee: New York University.
- **2025:** A Platelet Transcriptomic Signature of Thromboinflammation Predicts Cardiovascular Risk. *U.S. Provisional Patent Application No. 63/944,739*, filed December 19, 2025. With J.S. Berger, T.J. Barrett. Assignee: New York University.

PRESENTATIONS

- **2025:** Platelet Transcriptomic Signature Predicts Cardiovascular Risk in Peripheral Artery Disease. *Moderated Abstract Session, American Heart Association Scientific Sessions*
- **2025:** Proteomic Characterization of Chronic Coronary Disease. *NYU Department of Medicine Research Day*
- **2025:** Proteomic Characterization of Chronic Coronary Disease. *Poster Session, American College of Cardiology*
- **2024:** Clonal Hematopoiesis of Indeterminate Potential (CHIP) in Chronic Coronary Artery Disease: A Report from the ISCHEMIA Trials Biorepository. *Poster Session, American Heart Association Scientific Sessions*
- **2024:** A Transcriptomic Lupus Activity Signature Associates with Cardiovascular Function. *Poster Session, American College of Rheumatology Convergence*
- **2024:** Proteomic Characterization of Chronic Coronary Disease. *Cardiovascular Research Center Seminar Series, NYU CVRC*
- **2023:** Platelet Transcriptome in Type 2 Diabetes Provides Insight Into Peripheral Artery Disease and Cardiovascular Events. *Moderated Abstract Session, American Heart Association Scientific Sessions*
- **2023:** Predictive Modeling of a Genetic Platelet Activity Signature. *Poster Session, NYU Department of Medicine Research Day*
- **2023:** Predictive Modeling of a Genetic Platelet Activity Signature. *Poster Session, NYU CVRC Annual Symposium*
- **2022:** Multimodality Ultrasound Imaging to Investigate the Role of Von Willebrand Factor, Platelet Adhesion, and Oxidative Stress in Aortic Stenosis. *Moderated Poster Presentation, ASE Scientific Sessions, Seattle, Washington*
- **2022:** Augmentation of Pulmonary Perfusion by Conducted Effects of a Pulmonary Artery Ultrasound Catheter. *Poster Session, ASE Scientific Sessions, Seattle, Washington*
- **2021:** Catheter-based Therapeutic Ultrasound. *TOFS Research Conference, Knight Cardiovascular Institute, OHSU, Portland, Oregon*
- **2020:** Direct Vascular Effects of Catheter-based Ultrasound Devices Used For Facilitated Clot Lysis. *eAbstract Poster, American College of Cardiology*
- **2019:** Augmentation of Blood Flow by a High-Power High-Frequency Ultrasound Catheter. *Knight Cardiovascular Institute Research Retreat, OHSU, Portland, Oregon*
- **2017:** The Influence of Chemical Composition on Viscoelastic and Acoustic Properties of Microbubble Contrast Agents. *Thesis Seminar, Reed College, Portland, Oregon*

PUBLICATIONS

- [1] M. Muller, M. G. Cornwell, S. Rajkumar, Z. Chen, D. Coit, G. Drouard, P. Sastourne-Haletou, H. Yang, O. Raitakari, T. Lehtimäki, J. Hochman, D. J. Maron, J. S. Berger, J. D. Newman, K. V. Ruggles, and On behalf of the ISCHEMIA Biorepository Research Group, "Whole blood epigenomic and transcriptomic characterization identifies vulnerable molecular subtypes of chronic coronary disease," *Nature Communications*, June 2026.
- [2] C. Van Solingen, M. Muller, M. A. Sowa, and T. J. Barrett, "LUCAT1 Links Interferon Signaling in Platelets to Major Adverse Cardiovascular Events," *Arteriosclerosis, Thrombosis, and Vascular Biology*, p. ATVBAHA.125.324358, Apr. 2026.
- [3] C. A. Penfield, A. A. Robinson, A. Schaap, E. Luttrell-Williams, A. Hausvater, Y. Xia, M. Muller, M. S. House, V. Avtushka, L. Murphy, C. G. Zhou, J. S. Brandt, G. P. Quinn, A. S. Roman, D. R. Gossett, and J. S. Berger, "Platelet hyperreactivity and risk of ischemic placental disease: A prospective cohort study," *Pregnancy*, vol. 2, no. 2, p. e70255, 2026. <https://obgyn.onlinelibrary.wiley.com/doi/pdf/10.1002/pmf2.70255>.
- [4] R. Ni, M. Muller, K. V. Ruggles, T. J. Barrett, and J. S. Berger, "Whole-blood transcriptomics differentiates circulating gene expression between coronary artery disease and peripheral artery disease," *Vascular Medicine*, p. 1358863X251406528, Feb. 2026.
- [5] A. Beitzén-Heineke, M. A. Muller, Y. Xia, E. Luttrell-Williams, F. Schlamp, D. Voora, K. V. Ruggles, M. S. Garshick, T. J. Barrett, and J. S. Berger, "A platelet transcriptomic signature of thromboinflammation predicts cardiovascular risk," *JCI Insight*, vol. 10, Dec. 2025.
- [6] M. A. Muller, E. Luttrell-Williams, H. Bash, M. G. Cornwell, H. M. Belmont, P. Izmirly, H. Rosmann, M. S. Garshick, T. J. Barrett, S. Katz, K. V. Ruggles, J. P. Buyon, and J. S. Berger, "Platelet Gene Expression in Systemic Lupus Erythematosus and Cardiovascular Health," *JACC: Basic to Translational Science*, vol. 10, p. 101395, Dec. 2025.

- [7] C. E. Hamo, M. Muller, E. Rosenfeld, Y. Xia, A. Akinlonu, E. Luttrell-Williams, T. J. Barrett, and J. S. Berger, "Cardiometabolic risk factor burden associates with an immature platelet profile," *Platelets*, vol. 36, p. 2459800, Dec. 2025.
- [8] C. E. Hamo, M. A. Muller, J. D. Newman, J. Beckman, K. V. Ruggles, T. J. Barrett, and J. S. Berger, "Platelet reactivity expression score in diabetes mellitus," *Blood Vessels, Thrombosis & Hemostasis*, vol. 2, p. 100092, Nov. 2025.
- [9] M. S. Garshick, K. Drenkova, F. Kazatsker, I. Boothman, M. Muller, F. Schlamp, E. Luttrell-Williams, K. Lo Sicco, A. Neimann, J. U. Scher, B. Weber, J. M. Gelfand, J. Krueger, J. Buyon, and J. S. Berger, "Platelet Activation and a Platelet Biosignature Are Associated With Cardiovascular Risk in Patients With Controlled Psoriasis," *Arteriosclerosis, Thrombosis, and Vascular Biology*, vol. 45, pp. 2086–2096, Nov. 2025.
- [10] H. Sun, F. Schlamp, M. Muller, Y. Xia, S. Liberow, N. R. Smilowitz, J. S. Hochman, H. R. Reynolds, J. A. Beckman, T. J. Barrett, and J. S. Berger, "Platelets induce endothelial cell mitochondrial dysfunction in myocardial infarction," *Science Advances*, vol. 11, p. eadx0268, Nov. 2025.
- [11] S. P. Heffron, M. Muller, Y. Xia, E. Luttrell-Williams, C. B. Rockman, J. D. Newman, C. Rodriguez, T. J. Barrett, and J. S. Berger, "Low functional capacity in peripheral artery disease is associated with increased platelet activity and cardiovascular events," *Atherosclerosis*, vol. 405, p. 119200, June 2025.
- [12] M. Muller, R. Liu, F. Shah, J. Hu, C. Held, I. J. Kullo, B. McManus, L. Wallentin, L. K. Newby, M. S. Sidhu, S. Bangalore, H. R. Reynolds, J. S. Hochman, D. J. Maron, K. V. Ruggles, J. S. Berger, and J. D. Newman, "Clonal Hematopoiesis of Indeterminate Potential in Chronic Coronary Disease: A Report From the ISCHEMIA Trials Biorepository," *Circulation: Genomic and Precision Medicine*, vol. 0, no. 0, p. e004921, 2025.
- [13] T. J. Barrett, F. Schlamp, M. Muller, A. H. Lee, M. G. Cornwell, W. E. Luttrell, N. R. Smilowitz, J. Hochman, K. V. Ruggles, H. R. Reynolds, and J. S. Berger, "Myocardial Infarction Platelet Gene Expression Signatures in Women," *JACC: Basic to Translational Science*, vol. 10, pp. 307–322, Mar. 2025.
- [14] J. S. Berger, M. G. Cornwell, Y. Xia, M. A. Muller, N. R. Smilowitz, J. D. Newman, F. Schlamp, C. B. Rockman, K. V. Ruggles, D. Voora, J. S. Hochman, and T. J. Barrett, "A Platelet Reactivity Expression Score derived from patients with peripheral artery disease predicts cardiovascular risk," *Nature Communications*, vol. 15, p. 6902, Aug. 2024.
- [15] K. Ozawa, W. Packwood, M. A. Muller, Y. Qi, A. Xie, O. Varlamov, O. J. McCarty, D. Chung, J. A. López, and J. R. Lindner, "Removal of endothelial surface-associated von Willebrand factor suppresses accelerate atherosclerosis after myocardial infarction," *Journal of Translational Medicine*, vol. 22, p. 412, May 2024.
- [16] J. T. Belcik, A. Xie, M. Muller, and J. R. Lindner, "Influence of Atherosclerotic Risk Factors on the Effectiveness of Therapeutic Ultrasound Cavitation for Flow Augmentation," *Journal of the American Society of Echocardiography*, vol. 37, pp. 100–107, Jan. 2024.
- [17] K. Ozawa, W. Packwood, O. Varlamov, M. Muller, A. Xie, M. D. Wu, R.-J. Abraham-Fan, J. A. López, and J. R. Lindner, "Elevated LDL Cholesterol Increases Microvascular Endothelial VWF and Thromboinflammation After Myocardial Infarction," *Arteriosclerosis, Thrombosis, and Vascular Biology*, vol. 43, pp. 1041–1053, June 2023.
- [18] M. Muller, J. R. Lindner, and M. W. Hagen, "Ultrasound Imaging in Inflammation Research," in *Imaging Inflammation* (F. Man and S. J. Cleary, eds.), Progress in Inflammation Research, pp. 119–141, Cham: Springer International Publishing, 2023.
- [19] M. W. Hagen, S. Louey, S. M. Alaniz, T. Belcik, M. M. Muller, L. Brown, J. R. Lindner, and S. S. Jonker, "Changes in microvascular perfusion of heart and skeletal muscle in sheep around the time of birth," *Experimental Physiology*, vol. 108, no. 1, pp. 135–145, 2023. 1 citations (Crossref) [2024-02-21] eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1113/EP090809>.
- [20] M. A. Muller, J. Hodovan, K. Ozawa, M. W. Hagen, T. R. Hobbs, J. Templon, Y. Zhao, J. A. Kaufman, and J. R. Lindner, "Augmentation of Pulmonary Perfusion by Conducted Effects of a Pulmonary Artery Ultrasound Catheter," *Ultrasound in Medicine & Biology*, vol. 48, pp. 2146–2153, Oct. 2022. 1 citations (Crossref) [2024-02-21].
- [21] K. Ozawa, M. A. Muller, O. Varlamov, M. W. Hagen, W. Packwood, T. K. Morgan, A. Xie, C. S. López, D. Chung, J. Chen, J. A. López, and J. R. Lindner, "Reduced Proteolytic Cleavage of von Willebrand Factor Leads to Aortic Valve Stenosis and Load-Dependent Ventricular Remodeling," *JACC: Basic to Translational Science*, June 2022. 3 citations (Crossref) [2024-02-21].
- [22] T. C. L. Kohs, S. R. Olson, J. Pang, K. R. Jordan, T. J. Zheng, A. Xie, J. Hodovan, M. Muller, C. McArthur, J. Johnson, B. B. Sousa, M. Wallisch, P. Kievit, J. E. Aslan, J. D. Seixas, G. J. L. Bernardes, M. T. Hinds, J. R. Lindner, O. J. T. McCarty, C. Puy, and J. J. Shatzel, "Ibrutinib Inhibits BMX-Dependent Endothelial VCAM-1 Expression In Vitro and Pro-Atherosclerotic Endothelial Activation and Platelet Adhesion In Vivo," *Cellular and Molecular Bioengineering*, Apr. 2022. 5 citations (Crossref) [2024-02-21].
- [23] M. A. Muller, K. Ozawa, J. Hodovan, M. W. Hagen, D. S. H. Giraud, Y. Qi, A. Xie, T. R. Hobbs, P. S. Sheeran, and J. R. Lindner, "Treatment of Limb Ischemia with Conducted Effects of Catheter-Based Endovascular Ultrasound," *Ultrasound in Medicine & Biology*, Apr. 2021. 2 citations (Crossref) [2024-02-21].
- [24] W. Shentu, K. Ozawa, T. A. Nguyen, M. D. Wu, W. Packwood, A. Xie, M. A. Muller, E. Brown, M. W. Hagen, J. A. López, and J. R. Lindner, "Echocardiographic Molecular Imaging of the Effect of Anticytokine Therapy for Atherosclerosis," *Journal of the American Society of Echocardiography*, vol. 34, pp. 433–442.e3, Apr. 2021. 6 citations (Crossref) [2024-02-21].

- [25] M. A. Muller, T. Belcik, J. Hodovan, K. Ozawa, E. Brown, J. Powers, P. S. Sheeran, and J. R. Lindner, "Augmentation of Tissue Perfusion with Contrast Ultrasound: Influence of Three-Dimensional Beam Geometry and Conducted Vasodilation," *Journal of the American Society of Echocardiography: Official Publication of the American Society of Echocardiography*, pp. S0894–7317(21)00125–5, Mar. 2021. 6 citations (Crossref) [2024-02-21].
- [26] M. A. Muller, A. Xie, Y. Qi, Y. Zhao, K. Ozawa, M. Noble-Vranish, and J. R. Lindner, "Regional and Conducted Vascular Effects of Endovascular Ultrasound Catheters," *Ultrasound in Medicine and Biology*, vol. 46, no. 9, pp. 2361–2369, 2020. 4 citations (Crossref) [2024-02-21].
- [27] K. Ozawa, M. A. Muller, O. Varlamov, H. Tavori, W. Packwood, P. A. Mueller, A. Xie, Z. Ruggeri, D. Chung, J. A. López, and J. R. Lindner, "Proteolysis of Von Willebrand Factor Influences Inflammatory Endothelial Activation and Vascular Compliance in Atherosclerosis," *JACC: Basic to Translational Science*, vol. 5, pp. 1017–1028, Oct. 2020. 9 citations (Crossref) [2024-02-21].
- [28] H. Englander, A. Patten, R. Lockard, M. Muller, and J. Gregg, "Spreading Addictions Care Across Oregon's Rural and Community Hospitals: Mixed-Methods Evaluation of an Interprofessional Telementoring ECHO Program," *Journal of General Internal Medicine*, Sept. 2020. 14 citations (Crossref) [2024-02-21].
- [29] O. R. Mason, B. P. Davidson, P. Sheeran, Matthew Muller, J. M. Hodovan, J. Sutton, J. Powers, and J. R. Lindner, "Augmentation of Tissue Perfusion in Patients With Peripheral Artery Disease Using Microbubble Cavitation," *JACC: Cardiovascular Imaging*, vol. 13, pp. 641–651, Aug. 2020. 22 citations (Crossref) [2024-02-21].
- [30] B. P. Davidson, J. Hodovan, O. R. Mason, F. Moccetti, A. Gupta, M. Muller, J. T. Belcik, B. H. Annex, and J. R. Lindner, "Limb Perfusion During Exercise Assessed by Contrast Ultrasound Varies According to Symptom Severity in Patients with Peripheral Artery Disease," *Journal of the American Society of Echocardiography*, vol. 32, no. 9, pp. 1086–1094.e3, 2019. 9 citations (Crossref) [2024-02-21].